Development - phase 2.1

Thursday, 28 May 2026 8:06 PM

Current Design Review

GOOD THINGS

You already separated:

prototype-docs
from:

science-project generation
This is correct.
You are also tracking:

- photos
- notes
- cost sheets
- records

which is VERY important later.

🚀 Suggested Improvements

🟢 1️⃣☐ Rename Main Domain

Instead of:

prototype-docs
Better:

prototype-management
OR

prototype-execution
Reason:
Later it will include:

- workflows
- build status
- prototype lifecycle
- approvals
- validations

not just docs.

🟢 2️⃣☐ Improve Endpoint Hierarchy

Current:

/api/v1/prototypes/{prototype_id}/photos
This is GOOD.
Keep this style.

🚀 Recommended Structure

/api/v1/prototypes/

/api/v1/prototypes/{id}

/api/v1/prototypes/{id}/photos

/api/v1/prototypes/{id}/notes

/api/v1/prototypes/{id}/cost-sheets

/api/v1/prototypes/{id}/build-records

/api/v1/prototypes/{id}/materials

/api/v1/prototypes/{id}/status

🟢 3️⃣ Small Naming Improvements

Instead of:

records
Better:

build-records
because:

“records” becomes ambiguous later

🟢 4️⃣ Cost Sheet Endpoint

Current:

POST /cost-sheet
Better:

POST /cost-sheets
GET /cost-sheets
Reason:
Later:

- multiple revisions
- material comparisons
- pricing experiments

may happen.

🟢 5️⃣ Add Prototype Status Early

🟢 5️⃣☐ Add Prototype Status Early

VERY important.
Add:

/status
Possible values:

- planned
- material_pending
- in_progress
- completed
- failed
- needs_revision

This becomes useful VERY quickly.

🟢 6️⃣☐ Add Materials Endpoint Early

Because later:

- material analytics
- cheapest sourcing
- reusable inventory

becomes valuable.
Add:

/materials

🟢 7️⃣☐ Future Important Insight

Eventually:

science-project
and:

prototype
are DIFFERENT entities.

Example

Science Project

Hydraulic Bridge

Prototype Builds

Could have:

- cardboard version
- wood version
- low-cost version
- festival demo version

So:

one project can have multiple prototypes

VERY important future distinction.

🚀 Suggested Future Data Relationship

```
ScienceProject
  └── Prototype
        ├── Photos
        ├── Notes
        ├── CostSheets
        ├── BuildRecords
      └── Materials
```

🚀 MOST IMPORTANT THING

You are currently building:

execution intelligence

not just:

AI text generation

That distinction is VERY powerful long-term.

🚀 FINAL VERDICT

Your direction:

✅ Correct

Your hierarchy:

✅ Mostly correct

Improvements recommended:

- rename records → build-records
- pluralize cost-sheets
- add status endpoint
- add materials endpoint
- think project vs prototype separately

That will make your architecture MUCH cleaner later.
what is the purspose can you explain not getting it
You are currently thinking:

```
AI generates project → done
```

But your REAL system is actually:

```
AI generates project
        ↓
Human builds real prototype
        ↓
System captures execution intelligence
        ↓
Future generations improve
```

That second part is why these endpoints exist.

🚀 Let Me Explain With Real Example

Suppose AI generated:

Hydraulic Bridge Model
AI says:

- use syringes
- use cardboard
- use pipes
- estimated cost ₹250

Now your assistant actually tries building it.

🟢 What Happens In Real Life?

Assistant discovers:

- cardboard bends
- pipes leak
- cost became ₹480
- build took 2 days
- glue was insufficient
- children struggled assembling it

THIS is:

real execution intelligence

AI did not know this.
Your system must capture this knowledge.

🚀 WHY EACH ENDPOINT EXISTS

🟢 1️⃣ Photos Endpoint

POST /photos
Purpose:
Store:

- final prototype images
- build progress images
- proof prototype was actually built

Why important?
Later:

- teachers see examples
- students understand better
- demo showcases become possible
- future AI can reference successful builds

🟢 2️⃣ Notes Endpoint

POST /notes
Purpose:
Capture:

- build difficulties
- practical issues

- improvements
- observations

Example note:

Cardboard became unstable after water exposure

This becomes:

practical learning database

🟢 3️⃣ Build Records Endpoint

GET /build-records

Purpose:

Track:

- who built it
- when built
- build duration
- build attempts
- success/failure

Why?

Later you may have:

- multiple assistants
- workshops
- students building projects

Need operational tracking.

🟢 4️⃣ Cost Sheet Endpoint

POST /cost-sheets

Purpose:

Track:

- actual material costs
- travel costs
- build costs
- time investment

Why important?

Because later:

feasibility matters

Example:

AI says:

₹200 project

Reality:

₹950 project

Now system learns:

- unrealistic outputs
- expensive materials
- optimization opportunities

🟢 5️⃣ Materials Endpoint (Suggested)

Purpose:
Store:

- actual materials used
- alternatives
- local market availability

Example:

Thermocol unavailable locally
→ assistant used foam board
VERY valuable later.

🟢 6️⃣ Status Endpoint (Suggested)

Purpose:
Track prototype lifecycle.
Example:

planned
→ in_progress
→ completed
→ failed
This becomes important when:

- many prototypes exist
- assistant works remotely
- you manage builds asynchronously

🚀 MOST IMPORTANT THING YOU ARE BUILDING

Not:

AI generated projects
You are building:

educational execution intelligence

VERY different.

🚀 Long-Term Strategic Value

Over time your system learns:

AI Assumption	Real World Learning
cheap project	actually expensive
easy build	children struggle
safe material	breaks easily
simple workflow	takes too much time

This becomes:

your proprietary educational knowledge

🚀 Why This Is Powerful?

🚀 Why This is Powerful?

Most AI systems stop here:

text generation
Your system continues into:

real-world execution validation
That is MUCH harder to replicate later.

🚀 SIMPLE ANALOGY

Think like this:

AI Layer

recipe generator

Prototype Layer

actual cooking experience
Without actual cooking:

- recipe quality never improves.

Same here.